

OBJECTIVE

1. Students of Semester IV of MCA may select a project related to any of the technologies in which they are interested. They are to carry out a software project as part of curriculum. At the end of the semester the students are to submit a written working project report to be forwarded to GGSIP University. The objective of this Academic Circular is to standardize the format of submission of the project report and to lay guidelines to conduct the project.

University Scheme for Project

2. As per the syllabi of MCA (Paper No MCA-202), students have to submit a project as their final dissertation comprising of 20 credits. The dissertation has two components, viz.

(a) External: **Project** (60 Marks), where a written report is to be submitted. It involves external viva and presentation of the working project.

(b) Internal: **Project** (40 Marks), it includes continuous evaluation from idea generation to final presentation.

Scope of the Project

3. It is partly the responsibility of the student to find a relevant topic for his/her project and decide it in consultation with the concerned Faculty. The project work shall be an application development in a language/platform that the student has learnt during the previous semesters or is comfortable in. It shall be an in-house development. **Each student is required to carry out the independent work and submit the report individually. ANY PREVIOUS WORK OR BORROWED REPORT WILL BE SUMMARILY REJECTED AND IN ALL CASES OF REJECTION THE WORK IS TO BE REPEATED AFRESH.**

Proposal

4. The project proposal should be about one/two page long and must be submitted in writing to your respective guide. The format of the proposal is attached as **Appendix-A**. **All students are required to submit the proposal as per the schedule.**

Final Report

5. The Guidelines for methodology to be adopted for conducting the project is attached as **Appendix-B**. The format of the project report is attached as **Appendix-C**. **All students are to adhere to these guidelines.**

FORMAT OF THE PROJECT PROPOSAL/SYNOPSIS

1. Title of the Project:

The title of the project is to be meaningful and must convey the tools/platform used, e.g.,
“Speech Recognition System in ASP.NET”

While selecting the language/platform for development students are advised to select it where they have strength.

2. Problem Definition:

Students are required to give brief description of the system that is proposed to be computerized (e.g., in case of Speech Recognition System, your understanding of this system must be explained in brief). In addition students are required to give the exact outcome of the system. i.e., what/which part of the system is proposed to be computerized in the time schedule and available expertise?

3. Objectives & Scope: Explain the objectives and the scope of the project.

4. Methodology:

- (a) Explain the methodology for data collection.
- (b) Explain the technique (s) & tools/platform proposed to be used for systems analysis, design, testing and development of software.

5. Nature of Research: If you are writing the dissertation based on literature survey, give details of literature to be surveyed. Explain whether the nature of research is exploratory or original thought process is being pursued.

NOTE: ONCE THE TITLE, LANGUAGE/PLATFORM ARE FIXED THEY CANNOT BE CHANGED.

METHODOLOGY FOR PROJECT WORK/DISSERTATION

Chapter-1: Introduction/Problem Definition

1. Following aspects need to be covered in the first chapter:
 - (a) **Brief Description of the Organization:** If the project work is being done outside the Institute, describe the organization, its nature of business, products, turnover, market position etc.
 - (b) **General Description of the System under Study:** Briefly explain how the existing information system achieving the task, which is under study.
 - (c) **User Requirements:** Explain in non-technical terms, why the existing information system is not able to meet the current tasks against the objectives/standards/benchmarks of the organization. Problems faced by the existing system.
 - (d) **Establish the Need of the New System:**
 - (i) Give justification, How & Why without the new system user requirements cannot be fulfilled.
 - (ii) How will the proposed solution enhance the efficiency, effectiveness, better control, security and achievement of objectives/standards/benchmarks?
 - (e) **Objectives of the Project:** Keeping in view the user requirements and needs, available time and expertise, lay down the objectives of the project. Clearly spell out what is proposed to be done in the project.
 - (f) **Methodology:** SDLC, Structured Analysis & Design, Object Oriented Design method or Prototype method. You may follow a mixed methodology and tools; however, you should clearly mention the steps and tools and techniques that you are going to follow in the current project with brief justification.
 - (g) **Data Required & Data Collection Method:** You should establish requirement of data and methodology/technique of data collection, e.g., interview, questionnaire, document analysis. If questionnaire is developed, attach a copy of the questionnaire as appendix.

Chapter-2: Systems Requirement Analysis

2. In this chapter students are required to establish the user requirements by doing the systems analysis of the existing system in vogue and by interacting with the management/staff of the organisation. In every organisation a system does exist in some rudimentary or manual form, if not computerised. The activities that are required to be carried out and to be described in the project are:

- (a) Identify the process (es).
- (b) Identify the Input to and Output of each process, e.g., for Rail Reservation System input is the form and output is a ticket.
- (c) Identify the data elements (fields) in each input and output.
- (d) Identify the procedures/rules/mathematical relationships used for converting input into output.
- (e) Identify the controls (for I/O & access), security needs, validation rules and codes used for data elements.
- (f) Identify deficiencies in existing system.

3. Having collected the data on the above aspects, analyse it and interact with the management for any changes in process (es), input, output, procedures, access and control rights, security needs and your suggestions for removing the deficiencies.

Chapter-3: Systems Design

4. **Physical Design:** Explain relationship between various components (processes, input, output & entities) of the system. Draw DFDs and other diagrams.

5. **Explain Processing Logic:** Using Programme Flow Charts, Algorithm, Decision Table/Tree or Pseudo codes.

6. **Interface Design:**

(a) **Output Design:** Screen & Print layouts, i.e., the format and contents (fields) to be included – along with the file to which it is linked.

(b) **Input (Form or Screen) Design:** For keying-in data. Give fields, format, codes to be used, validation (error detection) requirements & the file to which it is linked.

7. **Database & File Design:**

(a) **Database Files/Tables Design:** Depending on the DBMS/RDBMS package used, give contents of each file including Field Name, Type, Size & Description.

(b) **Programme Files or Query Design:** Give purpose and detailed working of each programme file or a Query and link it to the processing logic and Process/Input/Output.

(c) **Compose Data Element Dictionary:** List all the fields in all the files in alphabetical order in a tabular form and serially numbered them.

Data Element S No	FIELD NAME	TYPE& SIZE	FIELD DESCRI PTION	FILE NAME/No	PROCESS NAME/No
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Chapter-4: Systems Development

8. **Purpose:** To carry out the activities of writing actual programmes, their debugging, testing and validation. Following activities are to be carried out:

(a) **Programme Development:** Explain the language or package used for developing the programmes (both back-end and front-end).

(b) **Testing & Debugging:** Use *Past Data* to check whether the programmes work as intended by

(i) **Module Testing:** Individual programme testing.

(ii) **System Testing:** Integrated module Testing for the entire system.

(You may like to include the test reports in the project to show the errors, if any, and a write-up on their rectification).

(c) **Validation:** Check the system with *Real-time Data* for input, output, computational processing & controls.

9. Attach programme codes, input and output designs and outputs using the real data in this chapter. Codes may be attached as appendix.

Chapter-5: Systems Implementation

10. Following activities are required to be conducted:

- (a) **Acquisition:** Workout the memory requirement, disk space required for programme (s) and database (s). List out HW, SW and people resources required and indicate the cost of the system.
- (b) **Conversion:** Suggest change over procedures (Parallel, Direct or Partial Change over) with justification.
- (c) **Training Needs:** Establish the training needs of operating personnel, clerical/non-clerical staff, supervisors, and senior staff.
- (d) **Documentation:**
 - (i) Include Operation Manual (instructions for running programmes).
 - (ii) User Manual-Giving instructions & procedures for end-users (for data entry, output, help etc).

FORMAT OF THE PROJECT REPORT

Format

1. The format for compilation of final report is given below:

- (a) Title Page
- (b) Certificate
- (c) Acknowledgements
- (d) Synopsis/Executive Summary
- (e) Table of Contents
- (f) List of Tables
- (g) List of Figures
- (h) List of Symbols
- (j) Body of the Project Report
- (k) References/Bibliography
- (m) Appendices

NOTE: The Report must be hard binded with navy blue color and with silver printing on it.

Title Page

2. The format of the title page is attached as **Annexure-I**.

Certificate

3. The format of the certificate is attached as **Annexure-II**. A certificate by the student, guide and duly authenticated by the Director/HOD is to be attached.

Acknowledgement

4. In the “Acknowledgement” page, the writer recognizes his indebtedness for guidance and assistance by the guide and other members of the faculty. Courtesy demands that he/she also recognizes specific contributions by other persons or institutions such as libraries and research foundations. Acknowledgements should be expressed simply, tastefully, and tactfully **duly signed above the name**.

Synopsis/Executive Summary

5. A synopsis is a brief or condensed summary of the project for higher-level management positions. It should be about 3-4 pages in length. It should comprise *problem definition, brief description of the system, objectives & scope of the project, methodology and tools used, volume of work carried out, limitations, and directions for future development*, if any.

Contents

6. The format of ‘Table of Contents’ and list of Tables/Figures/Symbols is attached as Annexure-III.

Body of the Project Report: Guidelines for Project Report Writing

7. The guidelines for the Body of the Project Report (methodology) are detailed in **Appendix-B**. Following aspects must be adhered to:

(a) Page Size: Good quality white A4 size executive bond paper should be used for typing and duplication.

(b) Chapter/Para Numbering: The chapters are to be numbered as Chapter-1, Chapter-2 etc. The heading/title of the chapter is to appear below the chapter number in uppercase. Paragraphs are to be numbered as 1,2,3 etc in every chapter separately. Subparas are to be numbered as 1.1, 1.2, 1.3----, 2.1, 2.2, 2.3 etc. Sub-sub paras. Are to be numbered as 1.11, 1.12, 1.13, 2.11, 2.12, 2.13 etc.

(c) Page Specifications:

- (i) Left Margin : 1 inch
- (ii) Right Margin : 1 inch
- (iii) Top Margin : 1 inch
- (iv) Bottom Margin : 1 inch
- (v) Gutter : 0.50 inch

(d) Page Numbers: All text pages starting from Body of the Project Report as well as program source code listings should be numbered at the **bottom center** of the pages.

(e) Normal Body Text:

- (i) **Font Size:** 12, Times New Roman, Double Spacing, Single Side Writing.
- (ii) **Paragraphs Heading Font Size:** 12, Times New Roman, Underlined
- (iii) **Page/Title Font Size:** 14

(f) Table and Figure Number: Table and figure numbers are to be written at the bottom of the table/ figure as given below:

(i) Table No-1: File Design for Employee Record

(ii) Figure No-1: Data Flow Diagram

(g) Binding & Color Code of the Report:

- (i) Hard Bound Report
- (ii) Background of the cover page – DARK BLUE**
- (iii) Letters in Silver White**

References/Bibliography

8. Examples are given below:

1. D.L. Carney, J.I. Cochran, "The 5ESS Switching System: Architectural Overview," *AT&T Technical Journal*, vol. **64**, no. **6**, July-August 1985, pp. 1339-1356.
2. A. Stevens, *C++ Database Development*, MIS Press, New York, 1992, p. 34.
3. J. Martin, *Computer Database Organization*, Prentice-Hall, Englewood Cliffs, NJ, 1977, p. 53.

4. Landsberger, J. (n.d.). Citing Websites. In *Study Guides and Strategies*. Retrieved May 13, 2005, from <http://www.studygs.net/citation.htm>.

APPENDICES

1. The appendices are to be attached at the end of the report and to be numbered as Appendix-A, Appendix-B etc right justified at the top of the page. Below the word Appendix write in parenthesis “Refer Para No_”. The para number is to be the number in the body of text where the reference of appendix is given. An appendix may have annexure (s). If there are annexure, these are to be attached immediately after the said appendix. The annexures are to be numbered as Annexure-I, Annexure-II etc.

Title of The Project Report

(Font size = 18)

*Submitted in partial fulfillment of the requirements
for the award of the degree of*

Master of Computer Application (MCA)

to

Guru Gobind Singh Indraprastha University, Delhi

Supervisor:
(Guide Name)
Designation

Submitted by:
(Student name)
Roll No.:



**Banarsidas Chandiwala Institute of Information Technology,
New Delhi – 110019
Batch (20xx-20xx)
MCA-202**

Certificate

I, Mr./Ms. _____, Roll No. _____
certify that the Project Report/Dissertation (MCA-202) entitled
“ _____ ” is done by me and it is an authentic
work carried out by me at _____ (Name of the
organisation or of the Institute). The matter embodied in this project work has not
been submitted earlier for the award of any degree or diploma to the best of my
knowledge and belief.

Signature of the Student

Date:

Certified that the Project Report/Dissertation (MCA-202) entitled _____ ”
Done by Mr./Ms. _____, Roll No. _____, is
completed under my guidance.

Signature of the Guide

Date:

Name of the Guide:

Designation:

Address:

Banarsidas Chandiwal
Institute of Information
Technology, Kalkaji, New
Delhi-110019

Countersigned

Director

ACKNOWLEDGEMENT

In the “Acknowledgements” page, the writer recognizes his indebtedness for guidance and assistance of the thesis adviser and other members of the faculty. Courtesy demands that he also recognise specific contributions by other persons or institutions such as libraries and research foundations. Acknowledgements should be expressed simply, tastefully, and tactfully **duly singed above the name.**

(Signature)
Student Name:
Student Roll No:

FORMAT FOR CONTENTS & LIST OF TABLES/FIGURES/ SYMBOLS

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	Chapter-3: System requirement analysis 3.1: Requirement Analysis 3.2: Specific Requirements	
8	Chapter-4: System Design 4.1: Work Flow diagram 4.2: Data flow diagrams : Entity-Relationship diagram : Use case diagrams	

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	: Database & file design	
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	: Query Language	
10	Chapter-6: Systems Testing	
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14	Future Directions	
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LIST OF TABLES/LIST OF FIGURES

Similar tables (as shown above for Contents) are to be drawn for List of Tables and List of Figures on separate pages.

LIST OF SYMBOLS

Symbols that are used in the body of the text are to be listed in tabular form as given below:

S No	Symbol	Nomenclature & Meaning
1	Σ	Sigma (Summation)
2	kbps	Kilo bits per second